① srcML of lines 14–15 and 18–20

```
<for pos:start="14:5">[g]for [a]<control><expr><name>qubit</name></expr><range>in <expr><call><name>range</name>
    ↪<argument_list>(<argument><expr><name>n[b]</name><operator>-</operator><literal type="number">1</literal>
    ↪</expr></argument>)</argument_list></call></expr></range></control><block>:<block_content>
        <expr_stmt><expr><call><name><name>qc</name><operator>.</operator><name>cx[c]</name></name>
            ↪<argument_list>(<argument><expr><name>qubit[d]</name></expr></argument>, <argument><expr><name>qubit
            ↪</name><operator>+</operator><literal type="number">1</literal></expr></argument>)</argument_list>
            ↪</call></expr></expr_stmt>
</block_content></block></for>

<expr_stmt pos:start="18:1">[g]<expr><call><name><name>qc</name><operator>.</operator><name>measure[c]</name>
    ↪</name><argument_list>(<argument><expr><name>n[d]</name></expr></argument>, <argument><expr><name>n[e]</name>
    ↪</expr></argument>)</argument_list></call></expr></expr_stmt>
<with>[f]with <expr><call><name><name>qc</name><operator>.</operator><name>if_test[f]</name></name><argument_list>(
    ↪<argument><expr><tuple>(<expr><name><name>qc</name><operator>.</operator><name>clbits[e]</name><index>[<expr>
    ↪<name>n</name></expr>]</index></name></expr>, <expr><literal type="number">1</literal></expr>)</tuple></expr>
    ↪</argument>)</argument_list></call></expr><block>:<block_content>
    <expr_stmt><expr><call><name><name>qc</name><operator>.</operator><name>x[c]</name></name><argument_list>(
        ↪<argument><expr><literal type="number">0[d]</literal></expr></argument>)</argument_list></call></expr>
        ↪</expr_stmt>
</block_content></block></with>
```

a `for` → unrolled **b** range → iteration count, from line 4 **c** gate
d qubit argument, evaluated per iteration **e** classical bit, written by the measurement and read by the condition
f `if_test` → conditioned block **g** position → source line

② Trace

line	iteration	operation
12, 13	every	12 rotations
15	0, 0	cx q_0, q_1
15	0, 1	cx q_1, q_2
15	1, 0	cx q_0, q_1
15	1, 1	cx q_1, q_2
17	–	h q_3
18	–	measure $q_3 \rightarrow c_3$
20	–	x q_0 if c_3

③ Slice of q_0

